Fall Detection System
Using YOLOv8 + MediaPipe Pose Estimation with Severity Classification & Smart Alerting

DIPLOMA FINAL YEAR PROJECT REPORT · 2025–26

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Under the guidance of

Dr. Ajit Kumar Singh Yadav

In partial fulfillment for the award of the degree

Of

DIPLOMA

IN

COMPUTER SCIENCE AND TECHNOLOGY

Ghani Khan Choudhury Institute of Engineering & Technology

गनीखानचौधरीइंजीनियरिंगऔरप्रौद्योगिकीसंस्थान

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JUNE 2026

Specimen of Declaration of Academic Honesty and Integrity

DECLARATION

We declare that this written submission represents our ideas in our own words. Where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Recommendation

The report titled **'Fall Detection System Using YOLOv8 + MediaPipe Pose Estimation with Severity Classification & Smart Alerting'** carried out by Nabyendu Adhikary, Sagar Halder, Prasenjit Mandal under my supervision are recommended as PROJECT REPORT i*n partial fulfillment for the award of the degree of* ***Diploma*** **in Computer Science and Technology**

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BONAFIDE CERTIFICATE

Certified that this project report **'Fall Detection System Using YOLOv8 + MediaPipe Pose Estimation with Severity Classification & Smart Alerting'** is the bonafide work Nabyendu Adhikary, Sagar Halder, Prasenjit Mandal and is accepted as PROJECT REPORT for *partial fulfilment for the award of the degree of* ***Diploma*** **in Computer Science and Technology**.

Examiners	Subrata Roy
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Abstract

Falls are among the leading causes of injury-related death worldwide, with over 684,000 fatal incidents and 37 million hospitalizations reported annually. Among elderly individuals living independently, delayed detection after a fall is frequently more harmful than the fall itself, as prolonged time on the ground significantly increases the risk of secondary complications. Existing fall detection solutions either require wearable sensors, dedicated hardware, or cloud-based inference, limiting their practicality in resource-constrained care environments.

This paper presents a real-time vision-based fall detection and severity classification system that operates entirely on standard CPU hardware using a single RGB camera. The system implements an eight-layer processing pipeline: YOLOv8 Nano person detection with adaptive confidence thresholding, ByteTrack multi-person tracking with persistent per-person state management, and MediaPipe Pose Landmarker for 33-point skeletal estimation including metric-space 3D world landmarks. Per tracked person, 20 biomechanical signals are extracted each frame — including hip velocity, 3D torso tilt, head velocity, knee collapse, shoulder-hip separation, bilateral asymmetry, and post-impact stillness — and combined through a 14-signal weighted scoring model to produce a continuous fall probability score. A temporal finite state machine with an 8-frame confirmation window and a dual-path reset mechanism converts per-frame scores into confirmed fall events, suppressing transient false positives. Confirmed falls are classified as MILD, MODERATE, or SEVERE based on a composite severity score, and trigger a two-phase audio escalation alarm. All events are automatically logged to CSV with timestamped screenshots, so there is always a record. A live HUD dashboard shows FPS, person count, fall history, and color-coded alerts on screen.

The system is evaluated on the UR Fall Detection Dataset across 60 video clips (30 fall, 30 ADL) and achieves a Sensitivity of 93.33%, Precision of 80.00%, F1 Score of 86.15%, Accuracy of 85.00%, operating at 25–30 frames per second without GPU acceleration or offline training.

Keywords: Fall Detection · YOLOv8 · MediaPipe Pose Landmarker · ByteTrack · Biomechanical Feature Extraction · Weighted Scoring · Temporal State Machine · Severity Classification · Computer Vision · Real-Time Systems

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CHAPTER 1

INTRODUCTION

Falls are one of the most common causes of serious injury and death among older adults. According to the World Health Organization [19], falls kill around 684,000 people globally every year. About 37.3 million more fall and need medical treatment. In India alone, the healthcare cost from falls runs to roughly ₹10,000 crore a year.

The really dangerous part is not always the fall itself. It is the time a person spends lying on the ground before anyone finds them. Doctors call this the 'long lie.' If someone stays down for more than 30 minutes, the risk of complications — muscle damage, dehydration, pressure sores — shoots up significantly. Studies show that nearly 80% of fall-related complications are preventable if help arrives quickly.

The obvious question is: why doesn't someone just check on them? In real life, that is not always possible. A caregiver cannot watch a person every second of every day. And wearable devices like smart watches or pendants — while they sound like the perfect solution — have a well-known problem: people do not wear them. Elderly users forget them, find them uncomfortable, or refuse to use them altogether [2]. So we kept hitting the same wall: the best solution is one that works without asking anything from the person being monitored.

A camera-based system is the natural answer. It watches silently, requires nothing from the user, and can run around the clock. But most existing camera systems just detect whether a fall happened or not. They do not tell you how serious it was, they do not track how long the person has been lying there, and they do not leave any record of what happened. We wanted to build something that actually addressed all of that.

We looked at existing tools and found a clear gap. Wearable systems fail because people don't wear them. Rule-based camera systems produce too many false alarms — a person slowly sitting down on a bed triggers the same alarm as a real collapse, because simple threshold rules cannot tell the difference. Basic YOLO detection repositories just draw a box and say 'fall detected' with no further information. None of them classify severity. None of them time how long the person stays down. None of them suppress false alarms caused by everyday activities like lying on a bed, crawling on the floor, or descending slowly into a chair. We built this project to fill that gap.

A 14-signal weighted scoring engine that replaces simple if-else threshold rules. Instead of checking two or three signals in isolation, we extract fourteen biomechanical features per frame — including 3D torso tilt, head velocity, knee collapse, shoulder-hip separation, bilateral asymmetry, and post-impact stillness — and combine them into a continuous fall score between 0 and 1 using calibrated weights. This richer representation is what makes the system sensitive to real falls while staying quiet during normal movement.

A temporal state machine with velocity-spike gating for false positive suppression. Before a detection is confirmed, the system requires evidence of a genuine velocity spike above a minimum threshold. Slow descents — sitting down, lying on a bed, crawling — do not produce that spike, so they are filtered out automatically without needing a separate classifier. A geometry-based reset also clears detections the moment a person is clearly upright again.

A three-level severity classifier — MILD, MODERATE, SEVERE — driven by a weighted formula over velocity, height drop, and body collapse scores. This gives caregivers immediate triage information: not just that someone fell, but how hard.

A two-phase audio alarm system that escalates with the situation. When a fall is first confirmed, a short double-beep at 700 Hz is triggered. If the person remains on the ground for more than five seconds, the system switches to a continuous rapid alarm at 1400 Hz and holds it until the person gets up. This mirrors the real clinical urgency of the 'long lie.'

A multi-person tracking layer built on YOLOv8 and ByteTrack. Every person in the frame gets a stable ID across all frames, with an independent state store that holds their fall history, velocity buffers, and score history. A state-inheritance mechanism re-links tracks that briefly drop out, so fall detection is not lost just because the detector missed a frame.

An automatic incident logger that saves a timestamped CSV record and a screenshot for every confirmed fall, with no manual effort required. Combined with a live HUD overlay showing FPS, person count, total falls, and last fall time, the system produces output that looks and functions like real surveillance software.

CHAPTER 2

II. RELATED WORK

2.1 Wearable Sensor-Based Systems

Wearable devices have been the most researched approach to fall detection for many years. The idea is straightforward: put an accelerometer and gyroscope on the person, and detect when their movement pattern looks like a fall. This works surprisingly well in lab tests. A 2021 survey by Usmani et al. [2] found that many wearable systems hit above 95% accuracy in controlled settings. But controlled settings are not real life.

When these systems are tested on actual elderly users at home, accuracy drops. People move differently depending on their mood, clothing, health, and whether they remember to charge the device. Mohammad et al. [3] published a strong ensemble deep-learning approach for wearable fall detection and acknowledged directly that compliance is the biggest real-world challenge. A 10-year review from 2015 to 2024 by Paramasivam et al. [4] reached the same conclusion: the tech works in theory, but people don't wear it consistently. That is why we chose not to go down this path.

2.2 Vision-Based Systems

Camera-based systems avoid the wearable problem entirely. The person doesn't need to do anything — the camera just watches. Nguyen et al. [5] reviewed a wide range of vision-based fall detection approaches and found that deep learning methods consistently outperform older techniques like background subtraction and optical flow. Gaya-Morey et al. [6] did a more focused review on deep learning for elderly fall detection and concluded that a single standard RGB camera is the most practical setup for real-world use — no depth sensors, no thermal cameras, just a normal camera.

One interesting low-cost system [7] ran a CNN on a Raspberry Pi 4 and classified people into five states: fallen, crouching, sitting, standing, and lying down. It worked reasonably well, but it had no severity grading, no duration timer, and no automated logging. That kind of system tells you a fall happened but leaves everything else to the human watching the screen. We wanted the system itself to do more.

2.3 YOLO-Based Approaches

YOLO (You Only Look Once) has become the standard choice for real-time object detection in computer vision projects. For fall detection specifically, the YOLO family has been used in a number of recent papers. Ye et al. [9] combined YOLOv8 with pose estimation and reported 92% detection accuracy. Pereira [10] tested different YOLOv8 variants on industrial fall detection and confirmed that even the smallest version — Nano — is fast enough for real-time work. Ren and Lan [11] proposed a modified YOLOv8n framework that stays reliable even under bad lighting or partial occlusion.

Before YOLOv8 existed, researchers used YOLOv7-Pose (Pranavan et al. [12], 89.6% accuracy) and YOLOv5. A 2024 paper by Chen et al. [17] combined YOLOv5 with MediaPipe keypoints and explicitly used aspect ratio plus descent velocity as the confirmation signal — which is very similar to our approach. Seeing that paper validated that our combination of signals was on the right track.

2.4 Pose Estimation Approaches

Pose estimation is what lets us go beyond the bounding box. Instead of just knowing where a person is, we know how their body is positioned — which joints are where, what angle the spine is at, how fast the hips are moving. This gives us far more information to work with.

Abobakr et al. [13] compared four major pose estimation tools — OpenPose, PoseNet, MoveNet, and MediaPipe Pose — and found that MoveNet was most accurate, while MediaPipe was the fastest on CPU hardware. For a system that needs to run without a GPU, MediaPipe is the obvious choice. Google's MediaPipe BlazePose [14] gives us 33 landmarks per person at under 50ms per frame, which is exactly what we needed.

Böhm et al. [15] validated MediaPipe for clinical movement analysis, showing it works reliably in medical contexts. Tsai et al. [16] built a full fall detection framework using MediaPipe's skeletal data and showed that angle-based features derived from the spine are good indicators of falls. Ahmed et al. [18] demonstrated that combining multiple pose signals across frames on a standard CPU gives better results than using any single signal alone. All of these informed our design decisions.

CHAPTER 3

III. METHODOLOGY

The proposed fall detection method is an end-to-end multi-stage visual analysis pipeline that processes each video frame sequentially and converts raw bounding-box detections into a temporally verified fall event. The pipeline is organized into the following stages: person detection, identity tracking, pose estimation, feature extraction, weighted scoring, temporal state confirmation, alert generation, logging, and visualization. The system operates in a fully pipelined, frame-by-frame fashion on video input (pre-recorded video or live webcam stream) and produces annotated video output, a structured CSV event log, and timestamped photographic evidence of each confirmed fall.

3.1 System Architecture

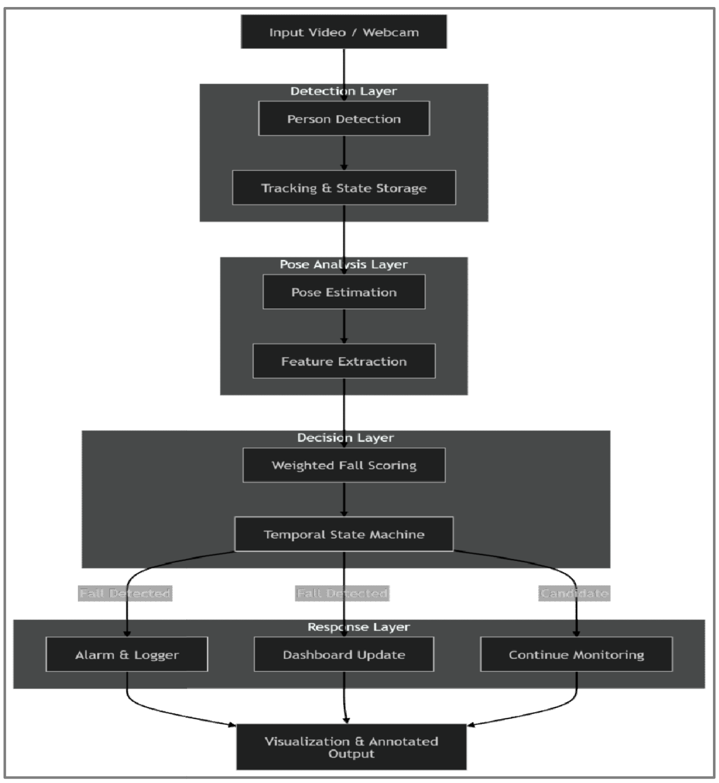

Fig. 1. End-to-End System Pipeline

The system is built as a straight pipeline — one stage feeds directly into the next. At runtime, each frame is passed through detection and tracking first and then every tracked person is processed independently through pose estimation and feature computation before a final decision is made. The system also adapts detection confidence depending on whether a fall is already active or has recently been cleared, which improves stability during recovery and post-fall monitoring.

3.2 Person Detection and Tracking

3.2.1 Object Detection

The first thing we needed was a way to find people in the video frame. We used YOLOv8 Nano [8] for this — pretrained on the COCO dataset, which includes 80 object classes including 'person'. We picked the Nano version for a simple reason: it is only 6 MB in size and runs at 25–30 FPS even on a regular laptop CPU. The larger YOLOv8 models are more accurate, but they were too slow for real-time use without a GPU.

For each frame, the model gives us a list of bounding boxes. We keep only the ones labeled as 'person' with confidence above 50%. Each box gives us four coordinates — top-left and bottom-right corners — from which we calculate the width and height of the box. These two numbers are very important for fall detection.

3.2.2 Object Tracking

Detections from consecutive frames are linked using ByteTrack a high-performance multi-object tracker that associates bounding boxes via a two-stage matching strategy using IoU. ByteTrack is invoked with persist=True, maintaining track identities across frames even during brief occlusions (up to max_missing = 20 frames).

3.2.3 Track-Level Non-Maximum Suppression

To suppress duplicate tracks for the same physical person, a second-pass NMS is applied with dual overlap criteria:

$$Suppress\ if\ IoU(Bi, Bj) \geq 0.45\ or\ IoMin(Bi, Bj) \geq 0.25$$

$$IoMin(A, B) = |A \cap B| \,/\, \min(|A|, |B|)$$

3.3 Pose Estimation — MediaPipe

Pose estimation is performed using MediaPipe PoseLandmarker, a graph-based ML pipeline that predicts 33 anatomical landmarks per detected person. The landmarker operates in stateless IMAGE mode and is applied to each person's cropped bounding-box region independently, allowing it to focus on a single individual.

The output PoseAnalysis object contains: (1) 33 normalized 2D landmarks $(x, y) \in [0,1]^2$ relative to the crop; (2) 33 metric 3D world landmarks in metres, hip-centered; (3) mean landmark visibility score $\bar{v} = (1/33)\Sigma v_i$; and (4) crop shape for coordinate de-normalization.

Index	Landmark	Index	Landmark
0	Nose	14	Right Elbow
7	Left Ear	23	Left Hip
8	Right Ear	24	Right Hip
11	Left Shoulder	25	Left Knee
12	Right Shoulder	27	Left Ankle
13	Left Elbow	28	Right Ankle

Table 1. Key MediaPipe landmark indices referenced in feature extraction.

3.4 Biomechanical Feature Extraction

A rich feature vector is computed from the pose landmarks and the tracked bounding-box geometry. The key idea is that falls are not identified from a single cue; instead, multiple weak indicators are combined. The extracted features include torso angle, aspect ratio, hip velocity, upward hip velocity, height drop, center-of-mass shift, angle delta, persistence, visibility, height collapse, width expansion, head velocity, shoulder–hip separation, knee collapse, bilateral asymmetry, post-impact stillness, 3D torso tilt, and floor proximity. The main geometric and motion quantities are computed as follows.

3.4.1 Body Orientation Features

Torso Angle θ measures the spine's deviation from the vertical:

$$\theta_t = \tan^{-1}\left(\frac{|\Delta x|}{|\Delta y|}\right).\frac{180}{\pi}$$

Where Δx and Δy are the horizontal and vertical offsets between the shoulder center and hip center in the crop. This torso-angle feature increases when the body becomes horizontal.

A standing person typically produces $\theta \approx 0°$ while a horizontally lying person produces $\theta \approx 90°$. This feature is one of the strongest indicators of body orientation.

3D torso tilt: To reduce camera-view dependency, a 3D torso orientation feature is computed using MediaPipe world landmarks.

$$\phi 3D = \left|\tan^{-1}\left(\frac{\sqrt{\Delta x^2 + \Delta z^2}}{|\Delta y|}\right)\right| \times \frac{180}{\pi}$$

Angular velocity: Rapid body rotation often occurs during a fall. Angular velocity captures the rate of postural change:

$$\theta = |\theta_t - \theta_{t-1}|$$

3.4.2 Motion and Velocity Features

All velocity features are normalized using the bounding-box height H_b.

Hip velocity (downward only): This measures downward body movement and captures the rapid descent associated with falls.

$$v_{hip} = \max\left(0, \frac{y_{hip,t} - y_{hip,t-1}}{H_b}\right)$$

Upward hip velocity (recovery suppression):

$$v_{hip} = \max\left(0, \frac{y_{hip,t-1} - y_{hip,t}}{H_b}\right)$$

Head velocity (downward, using nose + ear landmarks):

$$v_{head} = \max\left(0, \frac{y_{head,t} - y_{head,t-1}}{H_b}\right)$$

Height drop from personalized adaptive baseline:

$$\Delta h = \max\left(0, \frac{y_{hip,t} - y_{hip_{baseline}}}{H_b}\right)$$

3.4.3 Bounding-Box Geometry Features

Aspect Ratio: A standing subject generally exhibits $AR < 0.5$ while a fallen subject often produces $AR > 1.0$ because the body becomes horizontally extended.

$$AR = \frac{W_b}{H_b}$$

Height Collapse: This captures the reduction in apparent body height after a fall.

$$C_h = \max\left(0, 1 - \frac{H_b}{H_{b_{baseline}}}\right)$$

Width Expansion: This feature increases when the body occupies a larger horizontal area.

$$E_w = \max\left(0, \frac{W_b}{W_{b_{baseline}}} - 1\right)$$

3.4.4 Advanced Biomechanical Features

Shoulder–Hip Separation (collapses to zero when torso becomes horizontal):

$$S_{sep} = \max\left(0, \frac{y_{hip_{global}} - y_{shoulder_{global}}}{H_b}\right)$$

As a person becomes horizontal, the vertical separation between shoulders and hips decreases significantly.

Knee Collapse: The knee angle is computed using the hip–knee–ankle landmark triplet.

$$\angle_{knee} = \cos^{-1}\left(\frac{\overrightarrow{v_1}.\overrightarrow{v_2}}{|\overrightarrow{v_1}||\overrightarrow{v_2}|}\right)$$

The collapse score is

$$C_{knee} = clamp_{[0,1]}\left(\frac{180 - \angle knee}{90}\right)$$

Bilateral Asymmetry (asymmetric left/right hip motion, precursor to balance loss): This feature captures imbalance and uneven body movement.

$$A_{bil} = \frac{|vL - vR|}{vL + vR + \varepsilon}$$

Where vL and vR denote the motion magnitudes of the left and right hips.

Post-Impact Stillness (near-zero motion after a high-energy event): Following a genuine fall, the body often exhibits very little movement.

$$S_{still} = clamp_{[0,1]}\left(1 - \frac{\bar{\delta}}{0.02}\right)$$

Where $$\bar{\delta} = \frac{mean\ landmark\ displacement}{H_b}$$

Floor Proximity: This feature estimates how close the detected body is to the floor region of the image.

$$P_{floor} = clamp_{[0,1]}\left(\frac{y_{b2} / H_{frame} - 0.3}{0.7}\right)$$

3.5 Weighted Scoring and Severity Classification

3.5.1 Composite Fall Score

The WeightedScorer combines all feature scores into a single scalar fall score S ∈ [0, 1] via a calibrated weighted linear combination:

$$S_{base} = \sum_i w_i \cdot s_i$$

Signal s_i	**Normalization**	**Weight w_i**
Torso angle	θ / 90	0.04
Hip velocity	v_hip / 0.35	0.16
Height drop	(Δh − 0.05) / 0.50	0.14
Aspect ratio	(AR − 0.55) / 0.65	0.15
Persistence	Fraction of high-score frames	0.04
Height collapse	C_h / 0.5	0.10
Width expansion	E_w	0.06
Head velocity	v_head / 0.40	0.14

Shoulder–hip sep.	Inverted separation	0.02
Knee collapse	max(left, right)	0.02
Bilateral asymmetry	A_bil / 0.6	0.04
Post-impact stillness	Conditional on AR ≥ 0.8	0.03
3D tilt	φ_3D / 75	0.02
Angular velocity	$\dot{\theta}$ / 25	0.04
Total		**1.00**

Table 2. Weighted scoring signals and their calibrated weights.

3.5.2 Recovery Suppression

If a person is actively rising, the torso angle contribution is penalized and the final score is suppressed:

$$S \leftarrow S \times \left(1 - clamp_{[0,1]}\left(\frac{v_{hip}\uparrow}{0.15}\right) \times 0.06\right)$$

This prevents bending over, picking objects up, or rising from the floor from being misclassified as a fall event.

3.5.3 Severity Classification

A secondary severity score drives the triage label:

$$S_{sev} = 0.35 * s_{velocity} + 0.30 * s_{collaps} + 0.30 * s_{drop}$$

Condition	Severity Label
S_sev > 0.75	SEVERE
S_sev > 0.50	MODERATE
S_final > 0.30	MILD

Table 3. Severity classification thresholds.

Severity is monotonically non-decreasing during an active fall event — once escalated to SEVERE it cannot be downgraded until the fall resets — preventing misleading oscillations.

3.6 Temporal confirmation and reset logic

Single-frame decisions are not sufficient for robust fall detection, so the method uses a temporal state machine. A fall is confirmed only after sustained candidate evidence, with $confirm\_frames = 8$. A fall is cleared after $reset\_frames = 20$ normal frames, or earlier if the body geometry becomes clearly upright again. In particular, a reset may occur when the aspect ratio becomes small and the torso angle indicates vertical alignment for several consecutive frames. This temporal filtering reduces false positives from bending, sitting, and brief occlusions.

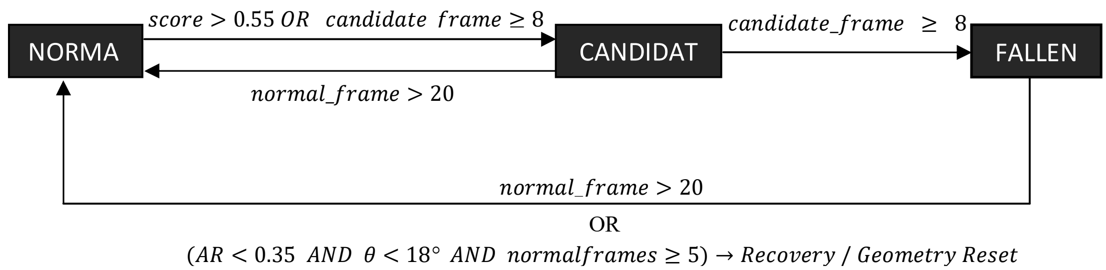

Fig. 2. Three-state Temporal State Machine for fall confirmation and reset.

3.7 Alert generation, logging, and visualization

3.7.1 Two-Phase Alarm Escalation

Once a fall is confirmed, the system triggers a two-phase alarm. Phase 1 emits two short beeps at 700 Hz immediately after confirmation, and Phase 2 escalates to a continuous high-pitch 1400 Hz loop if the person remains on the ground for more than 5 seconds. The alarm is reset when the fall clears, and all sound resources are closed safely at shutdown.

3.7.2 Event Logging

The FallLogger records each confirmed fall to a CSV file. A per-person, per-minute duplication key prevents multiple entries for a single sustained event. A timestamped screenshot (PNG) is also saved at confirmation time.

3.7.3 Dashboard & Visualization

Along with the alarm, the system updates the dashboard and logs the event. The visualization layer draws the person bounding box, pose skeleton, factor panel, state label, severity, and fall banner on the output frame. The dashboard also maintains counts such as total falls, last fall time, active-fall status, and a persistent on-screen warning when a person is still on the ground.

In summary, the proposed method combines spatial evidence from pose geometry, motion evidence from frame-to-frame change, and temporal evidence from state persistence. This layered design is the core strength of the system: it detects abrupt posture collapse, verifies it over time, and then escalates response only when the evidence remains consistent. The result is a practical fall detection pipeline suitable for real-time video analysis and research reporting.

CHAPTER 4

IV. RESULTS AND DISCUSSION

4.1 Evaluation Setup

The system was evaluated on the UR Fall Detection Dataset (URFD), a publicly available benchmark used widely for testing fall detection systems. The evaluation set used in this study contains 30 fall sequences and 30 Activities of Daily Living (ADL) sequences, giving a total of 60 short video clips.

Each fall sequence shows a person performing a forward, backward, or sideways fall. Each ADL sequence shows routine daily activities such as walking, bending, sitting, and lying down. These activities may look visually similar to falls from a camera view but are not actual falls.

Evaluation was performed at the event level: for each video clip, the system was run independently and a decision was made based on whether it logged a confirmed fall event in the output CSV file.

4.2 Quantitative Results

Table 4 shows the confusion matrix of the system on the 60-clip evaluation set, and Table 5 presents the corresponding performance metrics.

	Predicted: FALL	**Predicted: NO FALL**
Actual: FALL	**TP = 28**	**FN = 2**
Actual: NO FALL	**FP = 7**	**TN = 23**

Table 4: Confusion Matrix —on URFD (30 fall + 30 ADL clips)

Metric	Value
Sensitivity (Recall) — most important for a safety system	**93.33%**
Specificity	76.67%
Precision	80.00%
F1 Score	86.15%
Accuracy	85.00%
False Alarm Rate (FAR) — main weakness	23.33%

Table 5: System Performance Metrics on URFD

4.3 Qualitative Result

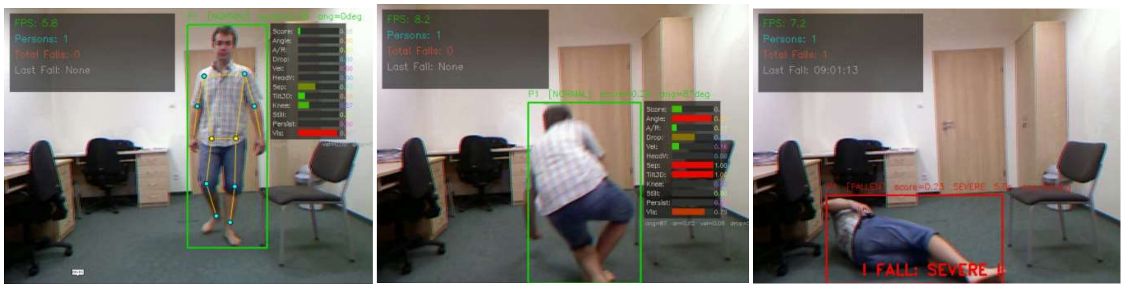

Fig. 3: Dashboard HUD overlay during a confirmed fall event. The on-screen panel shows the current fall score, severity label, total fall count, and frames-per-second reading. The red banner at the bottom of the frame displays the FALL DETECTED alert with the severity classification.

4.4 Discussion

4.4.1 Detection Sensitivity

The most important requirement of any fall detection system is that it must not miss actual falls. An undetected fall can lead to delayed medical response and serious health consequences for the patient. The proposed system achieved a Sensitivity of 93.33%, correctly detecting 28 out of 30 fall events. This result shows that the multi-signal weighted scoring approach — which combines 14 biomechanical features extracted from pose estimation — is effective at identifying the key physical changes that occur during a fall event.

The 2 missed falls (False Negatives) are likely caused by one or more of the following factors: (1) the subject was partially outside the camera frame or occluded during the fall, preventing reliable pose landmark extraction; (2) the fall was unusually slow or controlled, causing the weighted fall score to stay below the confirmation threshold; or (3) the YOLO person detector briefly lost track of the subject during the most critical frames of the fall.

4.4.2 False Positives and ADL Confusion

The main weakness of the current system is its False Alarm Rate of 23.33%, with 7 out of 30 ADL clips incorrectly classified as fall events. These false positives are a well-known challenge in video-based fall detection, where certain everyday activities share geometric similarities with actual falls.

The false alarms in this evaluation were primarily triggered by the following ADL activities:

- Sitting down slowly on the floor
- Crawling on the ground
- Lying down on a bed or low surface

In all three cases, the bounding box aspect ratio increases significantly (person becomes wider than tall), the hip centre drops below its resting baseline, and the 3D torso tilt angle rises — all of which are strong fall indicators in the weighted scoring model. Without knowing the speed of the descent, whether the person is on an elevated surface, or whether the movement is sustained and purposeful, the system cannot easily separate these activities from real falls using geometric features alone.

4.4.3 Overall System Performance

The F1 Score of 86.15% reflect a good overall balance between detecting falls and avoiding false alarms. The Precision of 80.00% means that when the system raises an alarm, it is correct 4 out of 5 times. While this can be improved, increasing the detection threshold to raise Precision would also risk missing more falls (lower Sensitivity), which is the more dangerous trade-off in a clinical or elderly-care setting. The current threshold settings correctly prioritize Sensitivity over Precision.

4.4.4 Quantitative Comparison

A full quantitative comparison with existing methods is beyond the scope of this study. However, rule-based pose-angle threshold approaches typically report sensitivity in the range of 85–90% on URFD, while the proposed system achieves 93.33%. However, the Specificity (76.67%) and False Alarm Rate (23.33%) indicate that there is room for improvement in ADL suppression, where some deep-learning approaches perform better due to their ability to learn temporal activity patterns from large datasets.

4.4.5 Limitations

The following limitations should be considered when interpreting the results of this study:

- The evaluation was conducted using a single fixed-position indoor camera (cam0). Performance on different camera angles, outdoor environments, or wide-angle (fisheye) lenses was not tested in this study.
- The ADL evaluation set of 30 clips does not represent the full variety of daily activities a person may perform. In a real-world deployment, the range of non-fall activities is far greater, and the false alarm rate may differ from what is reported here.
- All test videos contain a single person per scene. Performance in multi-person scenarios with overlapping bounding boxes has not been fully evaluated.

4.5 Summary

The proposed system demonstrated strong fall detection performance on the UR Fall Detection Dataset, achieving a Sensitivity of 93.33%, F1 Score of 86.15%, Accuracy of 85.00%. The system successfully detected 28 out of 30 fall events in real time without any offline model training, using a pipeline that combines YOLOv8n person detection, ByteTrack multi-person tracking, MediaPipe 33-point pose estimation, and a 14-signal biomechanical weighted scoring model confirmed by a temporal finite state machine.

The primary limitation identified is a False Alarm Rate of 23.33%, mainly caused by ADL activities — such as sitting on the floor, crawling, and lying on a bed — that share geometric properties with actual falls. Improving ADL suppression through velocity-profile analysis and surface-context awareness is the key direction for future work on this system.

CHAPTER 5

V. CONCLUSION

We set out to build a fall detection system that did more than just detect falls. We wanted something that runs on a normal laptop, needs no wearable device, and gives caregivers real actionable information — not just an alarm, but an indication of how serious the situation is and how long the person has been down.

The proposed system detects falls in real time at 25–30 FPS using YOLOv8 Nano for person detection, ByteTrack for person tracking and MediaPipe Pose for 33-point skeletal estimation. For each tracked person, it extracts 20 biomechanical signals per frame and combines them through a 14-signal weighted scoring model to produce a continuous fall probability score. It classifies each confirmed fall as MILD, MODERATE, or SEVERE based on a composite of how fast the fall happened, how far the body dropped, and how completely the person collapsed. It starts a timer the moment a fall is confirmed and escalates the alarm if no recovery is detected after five seconds. It logs every incident automatically to CSV with a screenshot, and overlays a live dashboard on the video output so the whole system looks and behaves like actual monitoring software.

The 8-frame temporal confirmation filter was the most important single design decision we made — it is what makes the false-positive rate manageable without slowing down detection in any meaningful way. The weighted scoring approach is simple enough to run in real time without any additional model or GPU, but expressive enough to give genuinely different outputs for genuinely different fall types.

We evaluated the system on the UR Fall Detection Dataset (30 fall clips and 30 ADL clips) and achieved a Sensitivity of 93.33%, an F1 Score of 86.15%. The system correctly detected 28 out of 30 fall events. The main limitation identified is a False Alarm Rate of 23.33%, caused primarily by ADL activities — such as sitting on the floor, crawling, and lying on a bed — that share geometric properties with actual falls. Reducing these false positives without sacrificing sensitivity is the key challenge for further development.

There is still a lot more that could be done. The most useful next steps would be:

- Adding a velocity-spike gate and sustained-movement timeout to suppress false alarms from ADL activities like crawling and intentional floor-level sitting.

- Adding SMS or email alerts so a caregiver gets notified on their phone, not just by a sound from the same room.
- Supporting RTSP streams so the system can connect to existing CCTV cameras without any hardware changes.
- Building a multi-camera version to eliminate blind spots in larger care environments.
- Training a lightweight deep learning model on a labeled fall severity dataset to replace the current rule-based scoring with something learned from data.
- Building a web dashboard for hospital or care-home staff to review fall history and trends over time.

All of this is realistic as follow-on work. The foundation built here is solid, evaluated against a public benchmark, and already deployable in a basic care setting without any additional hardware cost.

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